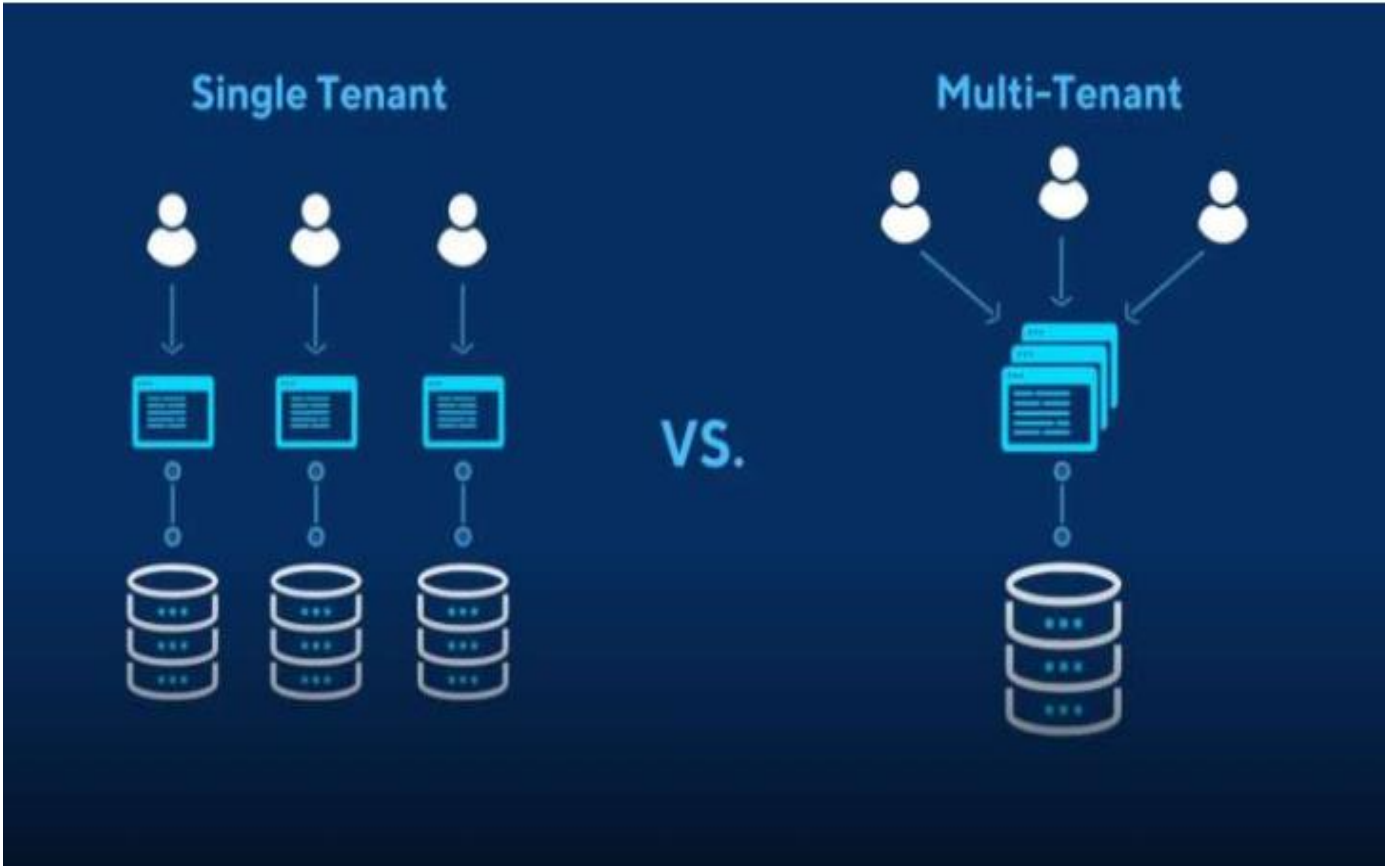
UNIT-I

Part-II

Cloud Concepts and Technologies

- The concepts and Technologies that are covered :

1. Virtualization
2. Load Balancing
3. Scalability and Elasticity
4. Deployment
5. Replication
6. Monitoring
7. Map Reduce
8. Identity and Access Management
9. Service Level Agreements
10. Billing


Single Tenant
VS.
Multi-Tenant

Virtualization

- Virtualization refers to the partitioning the resources of physical system into multiple virtual resources.
- Virtualization enables pooling**(grouping)** of resources to serve multiple users using multi-tenancy in cloud computing.
- Users are assigned virtual resources that run on top of the physical resources.

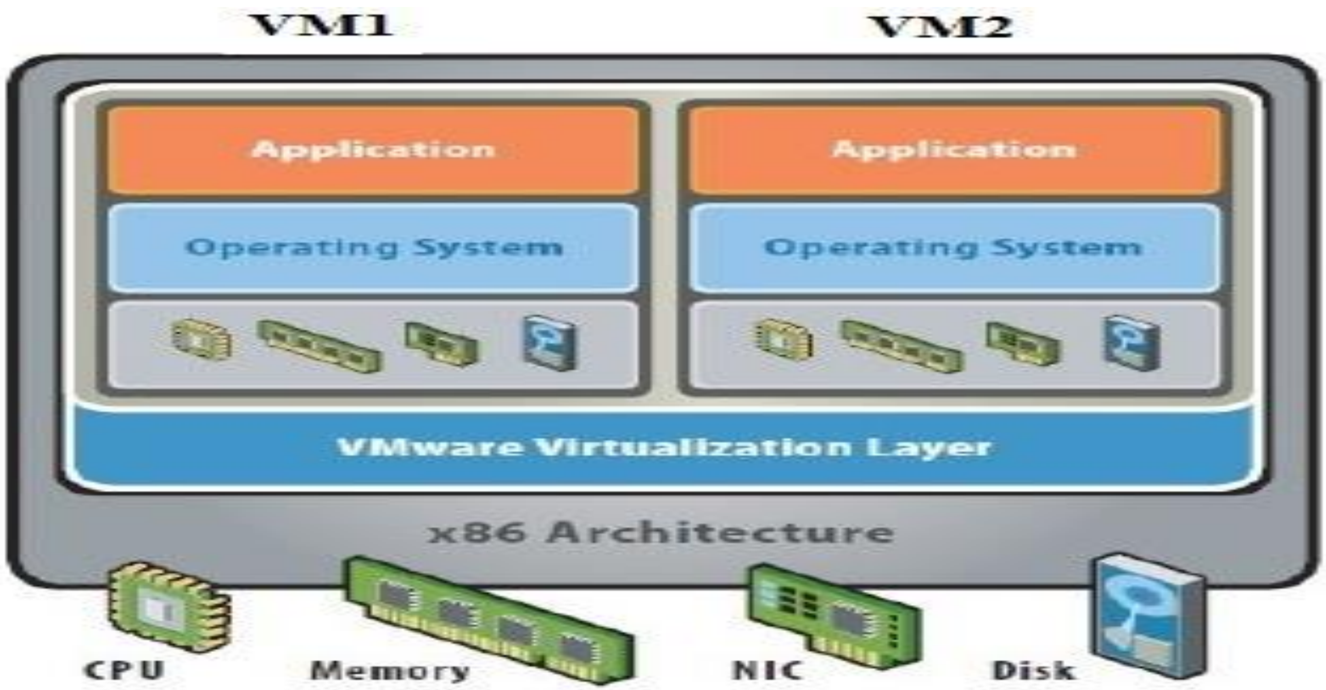


Fig : Virtualization Architecture

VIRTUALIZATION BASICS

Use a single physical machines' hardware
to run multiple virtual machines within it

Key Points:

- Use a system's hardware
- Allocate CPU/RAM/Storage to VMs
- Cannot exceed the CPU/RAM/Storage that is available on the physical hardware

Benefits:

- Better use of Hardware resources
- Power saving/reduced footprint
- Recovery (VMs can be saved as files)
- Flexibility (VMs can be moved)
- Researching Operating Systems & other Software (sandbox)

Virtualization

Hypervisor

- It is an Interface or a monitoring system in virtualization layer to present a **virtual operating platform** to a guest operating system.
- There are two types of hypervisors :

1. Type- 1 Hypervisor or Native Hypervisor

- In this Hypervisor**(Hypervisors make it possible to use more of a system's available resources)** run directly on the host hardware and control the hardware and **monitor the guest operating system**.

Ex : Citrix Xen Server,Oracle VM ,KVM,VMWare ESX/ESXi,Hyper-V.

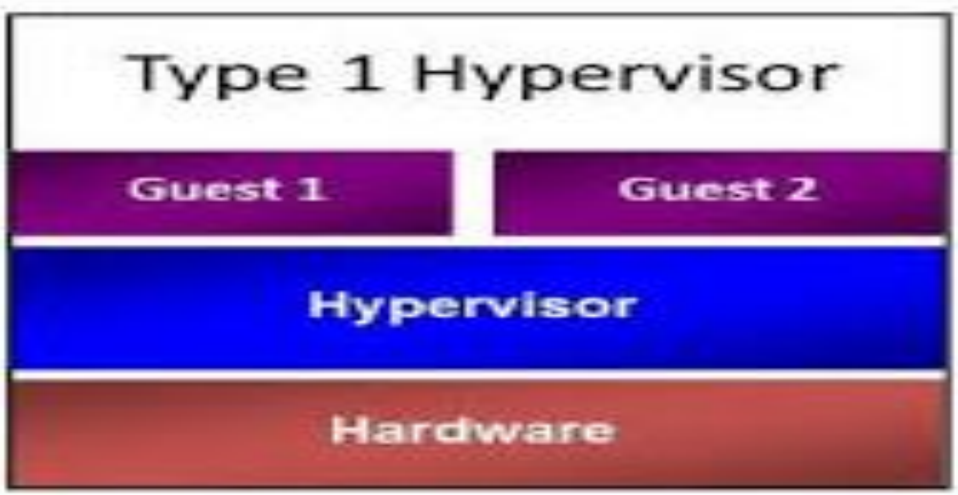


Fig : Type-1 Hypervisor

Virtualization

2. Type- 2 Hypervisor or Hosted Hypervisor

- Hypervisor run on the top of the conventional operating system and monitor the **guest operating system**.

Ex : VMWare Workstation ,Virtual Box.

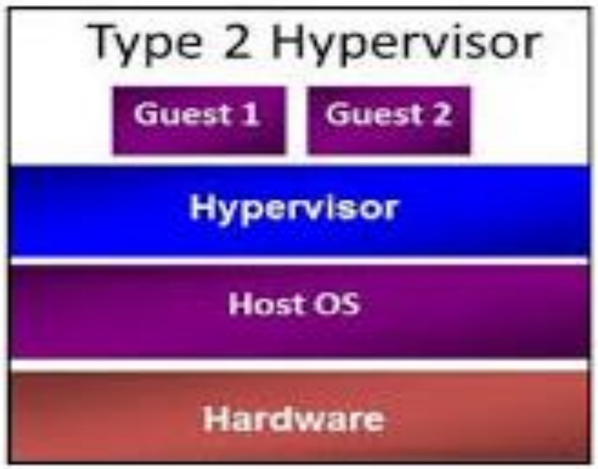


Fig : Type-2 Hypervisor

Guest OS

- It is an operating system that is installed in a virtual machine in addition to the host or main OS.

Virtualization

- There are different forms of virtualizations approaches :

1. Full Virtualization

- In this the **virtualization layer completely decouples** the guest OS from the underlying hardware .
- The guest OS requires no modification and is not aware that it is being virtualized.
- Full virtualization is enabled by direct execution of user requests and binary translation of OS requests.

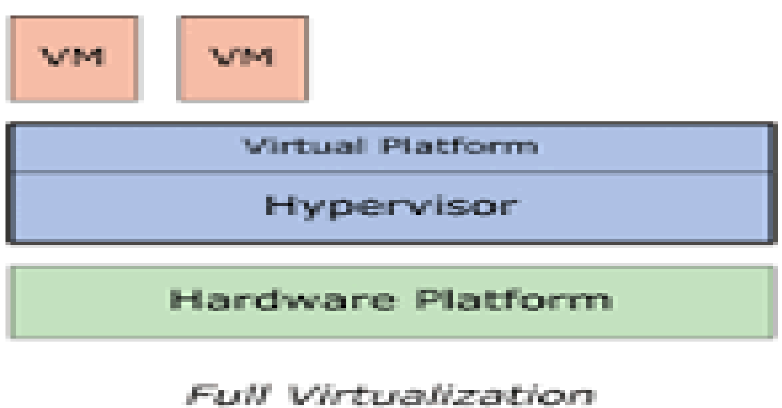


Fig : Full Virtualization

Virtualization

2. Para-Virtualization

- In this the guest operating system is modified to enable communication with the hypervisor to improve performance and efficiency.
- The guest OS kernel is modified to replace nonvirtualizable instructions with hypercalls that communicate directly with the virtualization layer hypervisor.

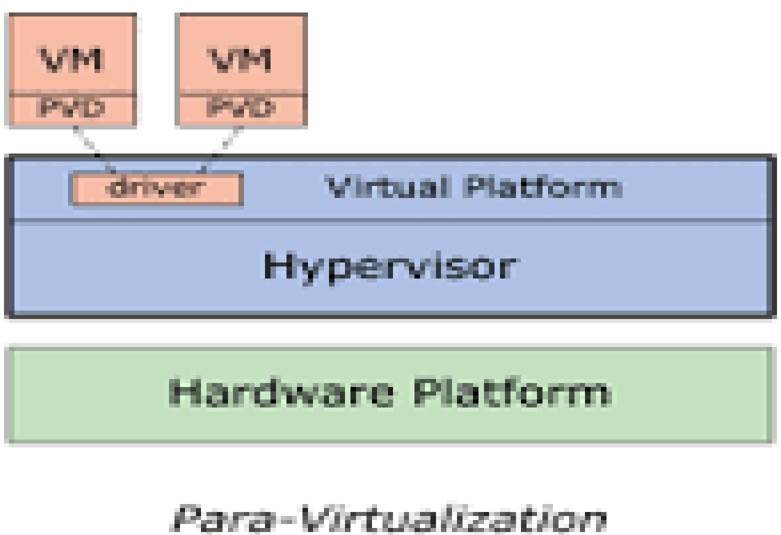


Fig : Para-virtualization

Virtualization

3. Hardware Virtualization

- Hardware assisted virtualization is enabled by hardware features such as Intel’s Virtualization Technology (VT-x) and AMD’s AMD-V.
- In this privileged and sensitive calls are set to automatically trap to the hypervisor .

Load Balancing

- Load balancing distributes workloads across multiple servers to meet the application workloads and to achieve one of the important feature of cloud computing scalability **(capacity to be changed).**
- The main goal of load balancing techniques are to achieve maximum utilization of resources, minimizing the response times, maximizing throughput.
- With load balancing, cloud based applications can achieve high availability and reliability.
- The routing of user requests is determined based on a load balancing algorithm.
- There are different types of load balancing algorithms are available. They are :

1. Round Robin 2. Weighted Round Robin 3. Low Latency
4. Least Connections 5. Priority 6. Overflow

Load Balancing

1. Round Robin

- In this load balancing , the servers are selected one by one to serve the **incoming requests** in a non-hierarchical circular fashion with no priority assigned to a specific server.

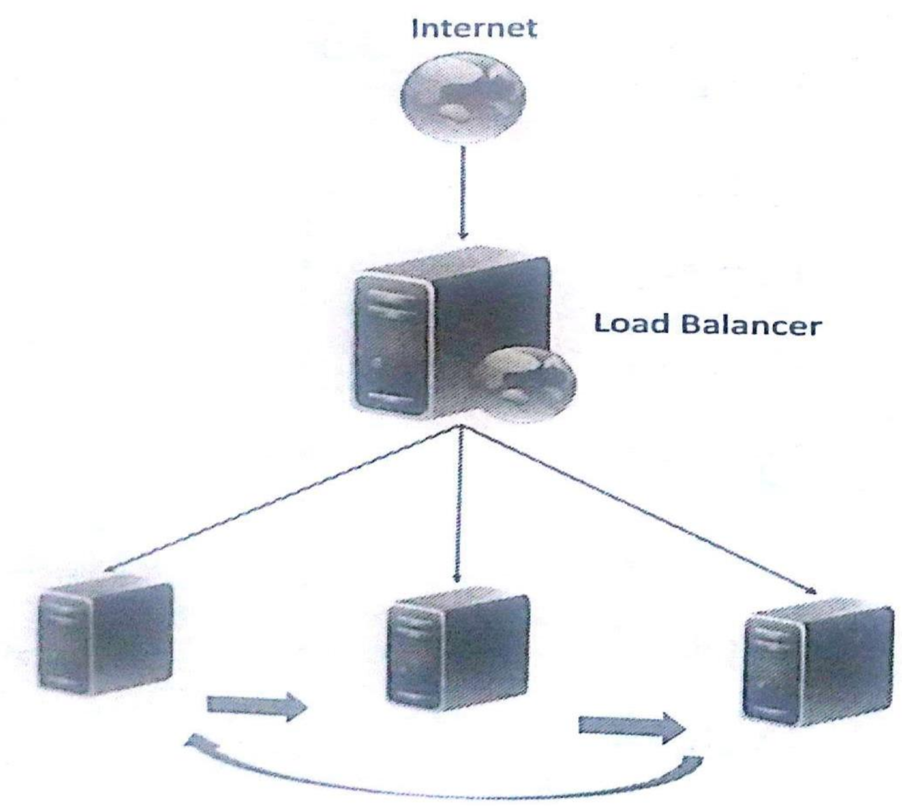


Fig : Round Robin

Load Balancing

2. Weighted Round Robin

- In weighted round robin load balancing, Servers are assigned some weights. The incoming requests are proportionally routed using a static or dynamic ratio of respective weights.

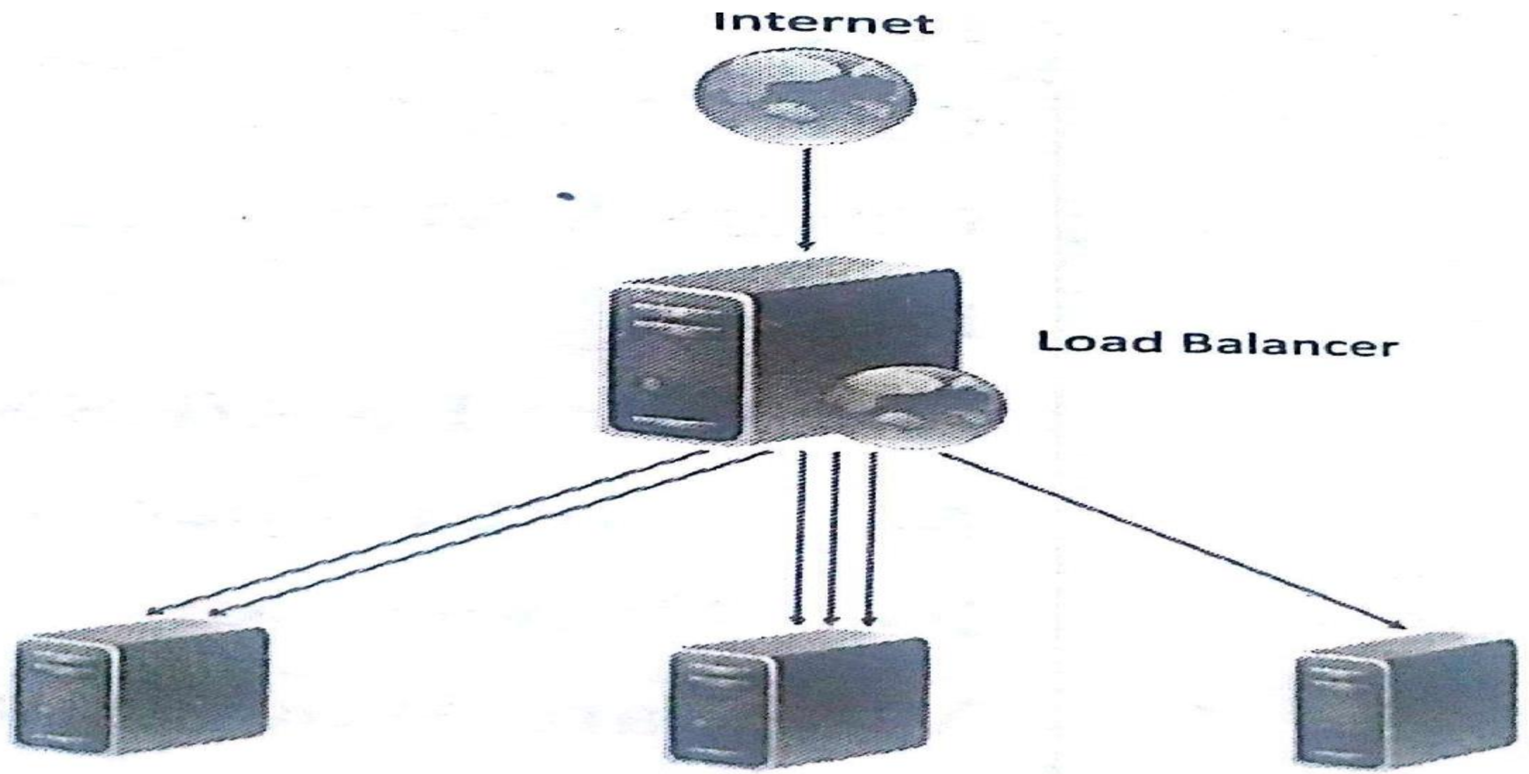


Fig : Weighted Round Robin

Load Balancing

3 . Low Latency

- In low latency load balancing the load balancer monitors the latency of each server. Each incoming request is routed to the server which has the **lowest latency(delay between a user's action).**

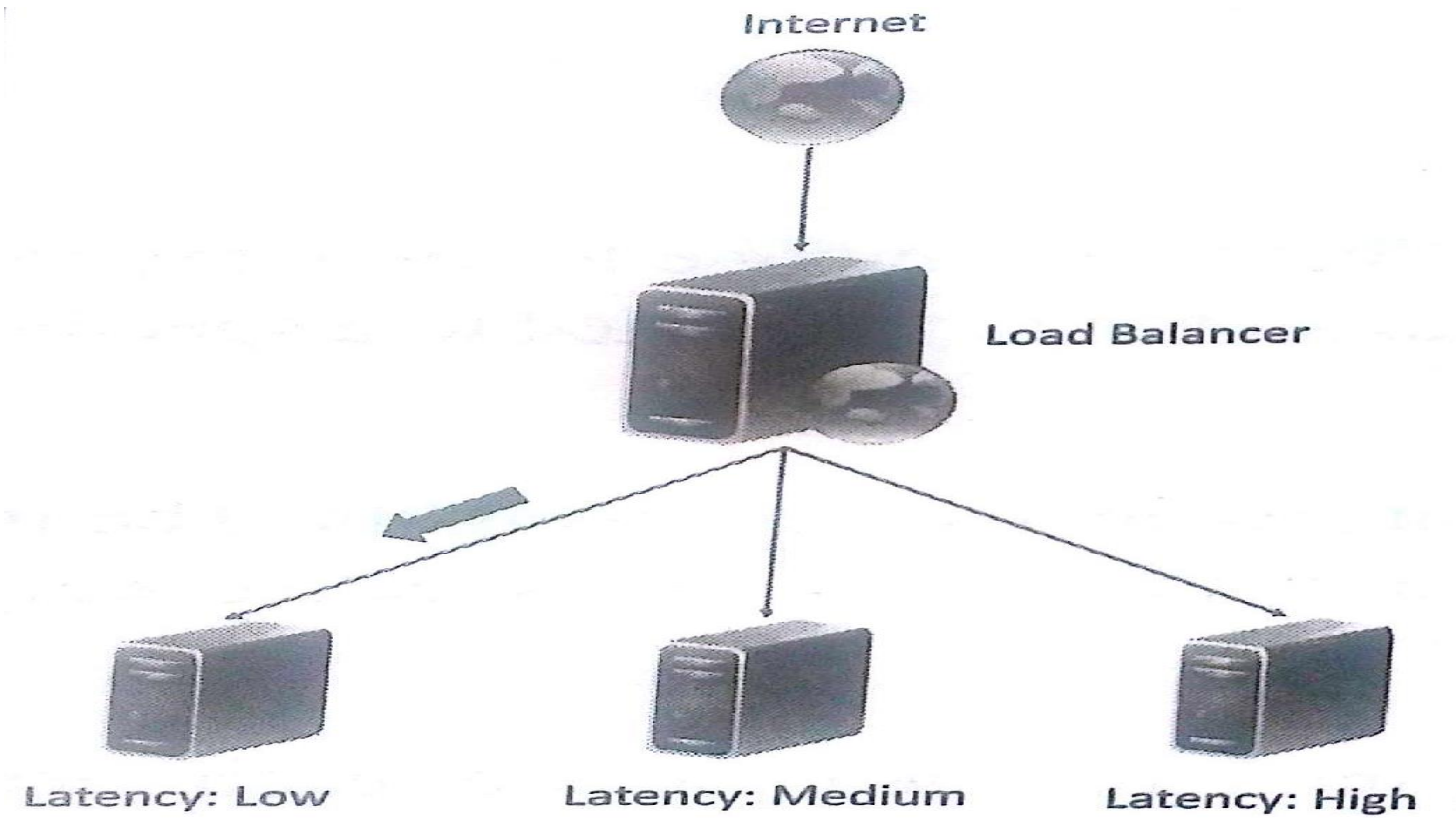


Fig : Low Latency

Load Balancing

4. Least Connections

- In least connection load balancing, the incoming requests are routed to the server with the **least number of connections.**

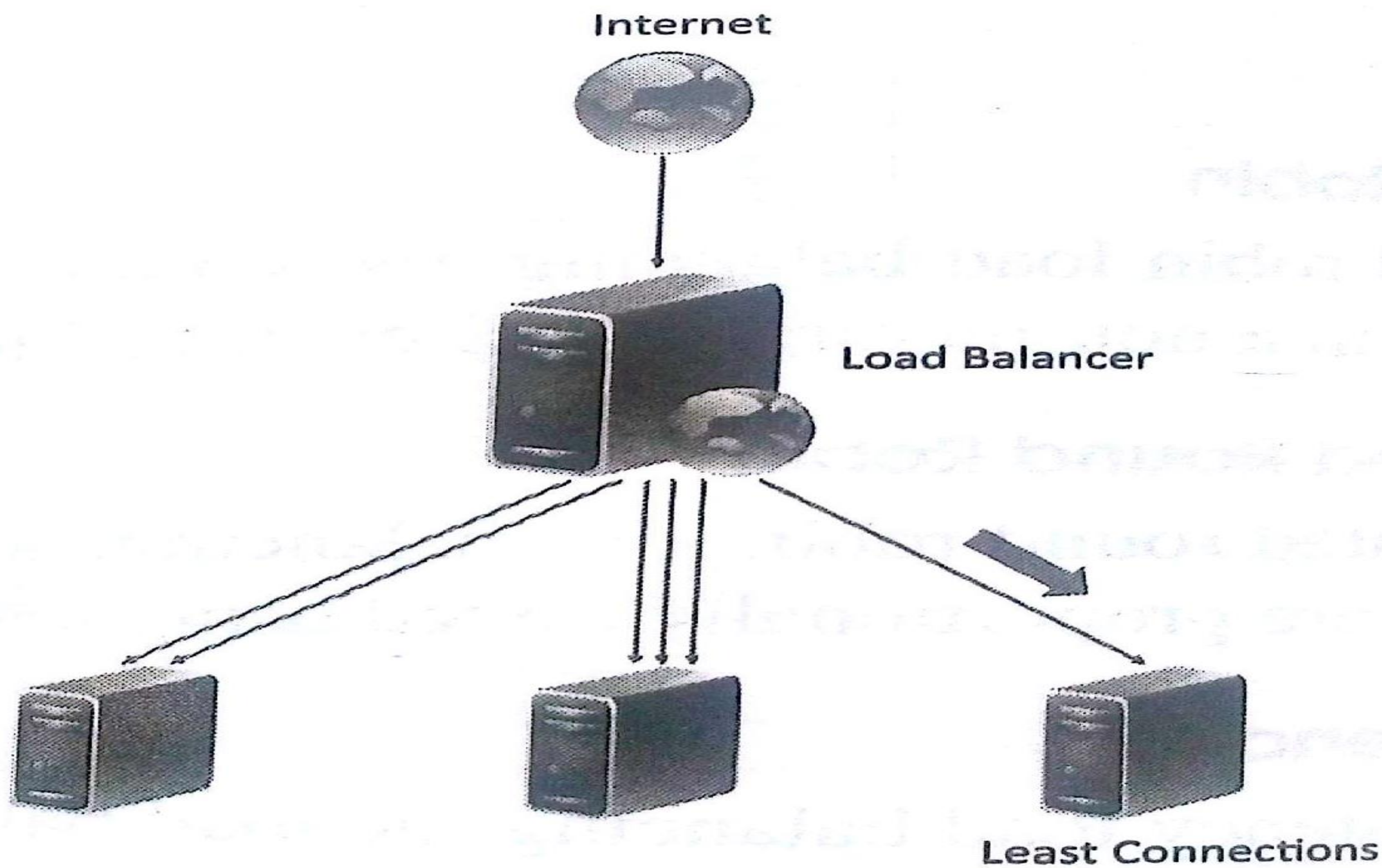


Fig : Least Connections

Load Balancing

5. Priority

- In priority load balancing, each server is assigned a priority.
- The incoming traffic is routed to the **highest priority server** as long as the server is available.
- When the highest priority server fails, the incoming traffic is routed to a server with a lower priority.

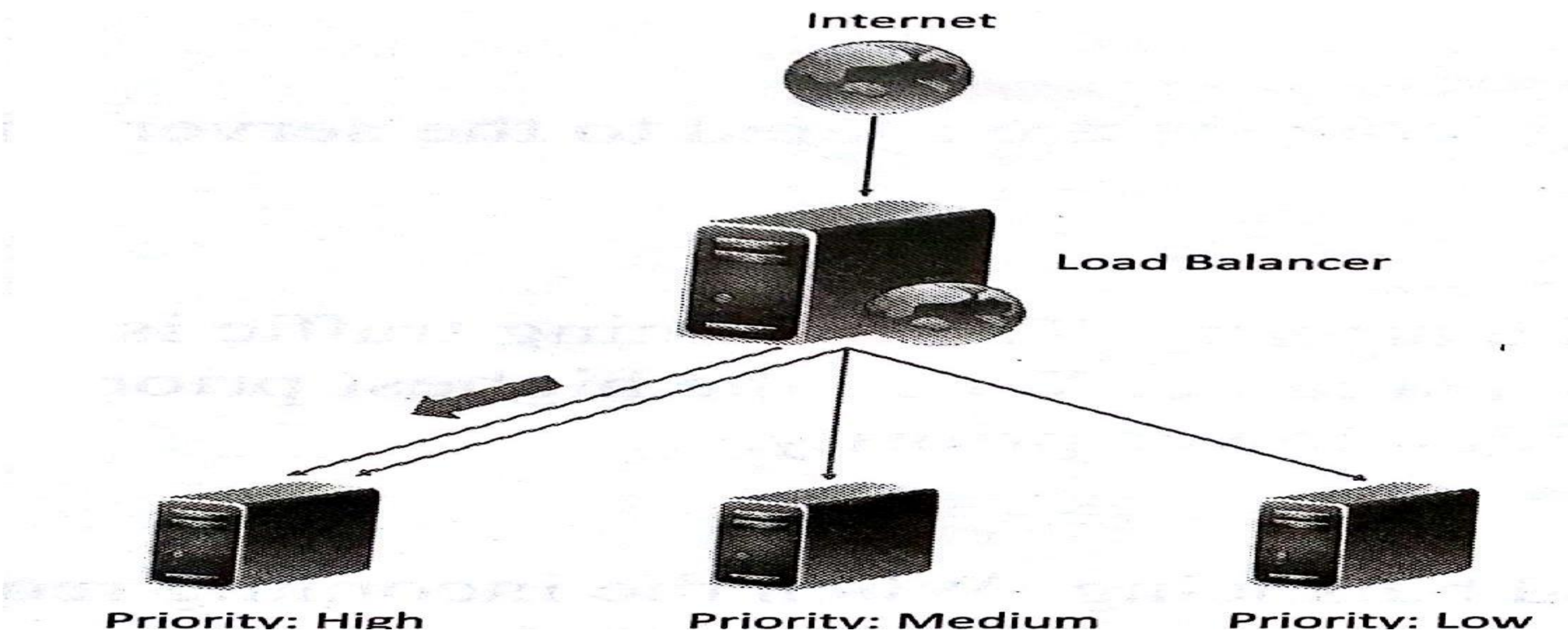


Fig : Priority

Load Balancing

6. Overflow

- Overflow load balancing is similar to priority load balancing .
- When the **incoming requests to highest priority** server overflow, the requests are routed to a lower priority server.

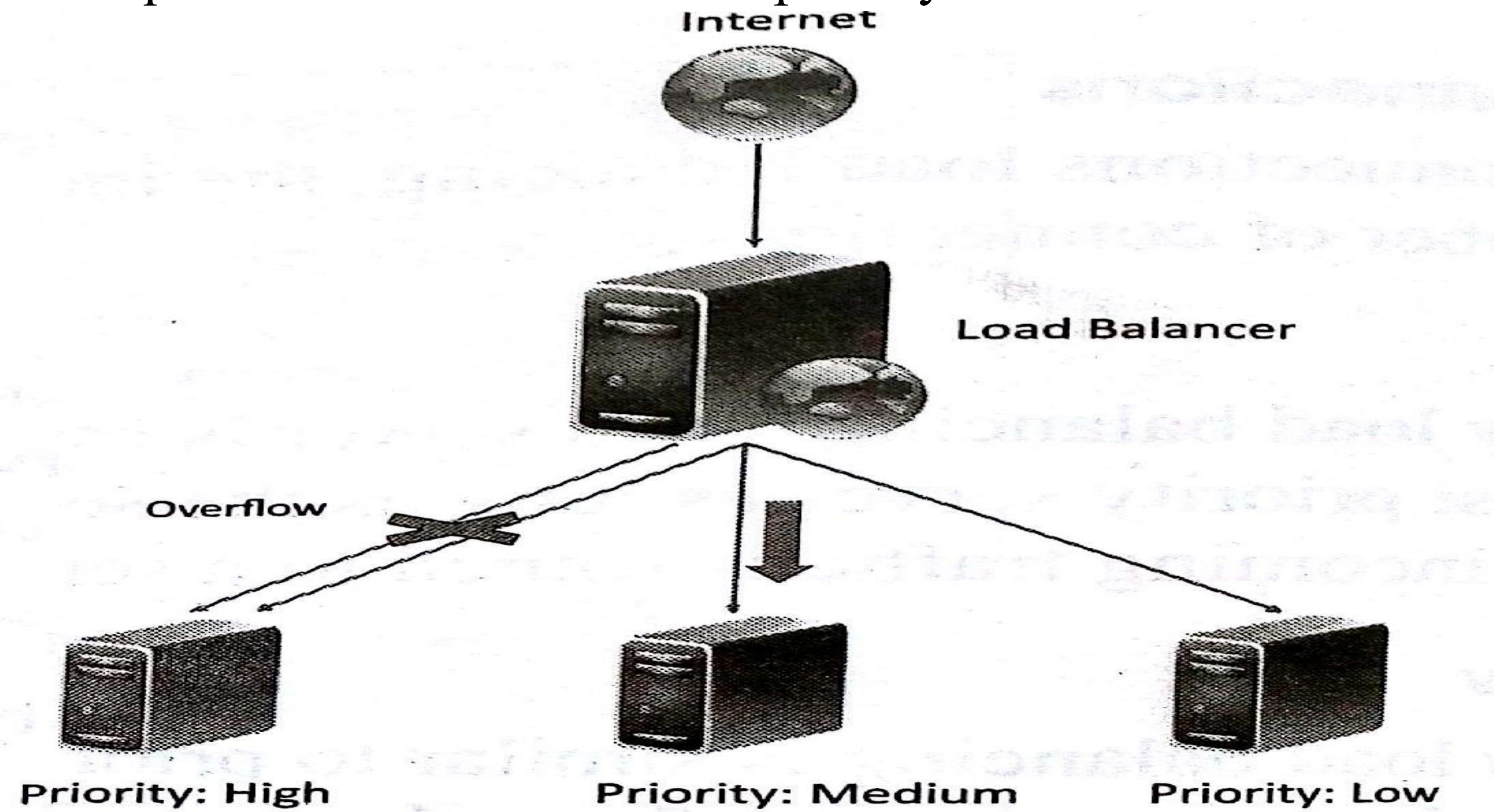


Fig : Overflow

Load Balancing

- An important issue in load balancing is for session based applications is maintaining the state or the information of the session is important.
- Commonly used persistence approaches are :

1. Sticky Session

- In this technique all the requests belonging to a user session are routed to the same server. These sessions are called sticky sessions.
- **Advantage** of this approach is that it makes session management simple.
- **Disadvantage** is that if a server fails all the sessions belonging to that server are lost, since there is no automatic fail over is possible.

Load Balancing

2. Session Database

- In this approach, all the session information is stored externally in a separate session database, which is often replicated**(reproduce)** to avoid a single point of failure.
- It allows automatic failover **(a procedure by which a system automatically transfers control to a duplicate system when it detects a fault or failure).**

3. Browser Cookies

- In this approach, the session **information is stored on the client side** in the form of browser cookies.
- The advantage of this approach is that it makes session management easy and **has the least amount of overhead for the load balancer.**

Load Balancing

4. URL re-writing

- In this technique, a URL**(Uniform Resource Locator)** re-writing engine stores the session information by modifying the URLs on the client side.
- A drawback is that amount of session information that can be stored is limited.
- Applications that require larger amounts of session information ,this approach does not work.
- Load balancing can be implemented in software or hardware.
- Software based load balancing can be run on standard operating system ,like other cloud resources , load balancers are also virtualized. Ex : Nginx, HAProxy, Pound, Varish.
- Hardware based load balancing algorithms are implemented in Application Specific Integrated Circuits(ASICs).

  Ex : Cisco Systems catalyst 6500, Coyote Point Equalizer, F5 Networks BIG-IP LTM, Barracuda Load Balancer.

Scalability(Capability) and Elasticity (Ability)

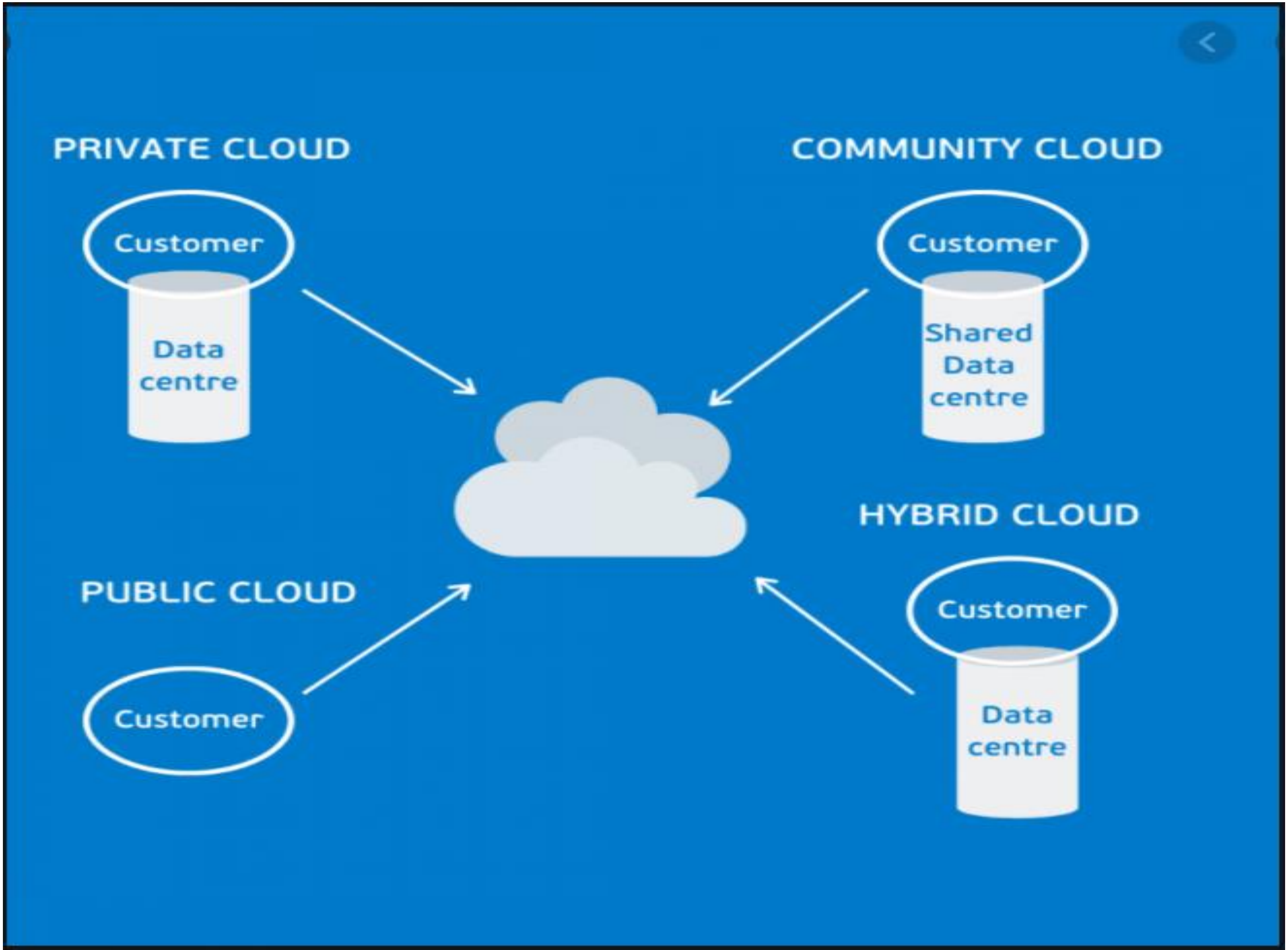
- Multi tier application like **e-commerce**, **social networking, business-to-business** etc. can experience rapid changes in their traffic.
- Capacity planning is an important task for the application that have multiple tiers of deployment with varying number of servers in each tier.
- Capacity planning involves determining the right sizing of each tier of the deployment of an application in terms of the number of resource and the capacity of each resource.
- Capacity planning may be for computing, storage, memory or network resources.
- Traditional approaches for capacity planning are based on predicted demands for applications and account for worst case peak loads of applications.
- When the work loads of application are increase, the traditional approaches have been either to scale up or scale out.

- Scaling up involves upgrading the hardware resources.
- Scaling out involves addition of more resources same type.
- Traditional scaling up and scaling out approaches are based on demand forecast at regular intervals.
- When variations in workloads are rapid, traditional approaches are unable to keep track with the demand and lead to either over-provisioning or under –provisioning of resources.
- Over-provisioning leads to high capital expenditure than required and under-provisioning leads to traffic overloads, slow **response times**, **low throughput** and hence loss of the opportunity to server the customers.

Scalability v/s Elasticity

Scalability	Elasticity
"Increasing" the capacity to meet the "increasing" workload.	"Increasing or reducing" the capacity to meet the "increasing or reducing" workload.
In a scaling environment, the available resources may exceed to meet the "future demands".	In the elastic environment, the available resources match the "current demands" as closely as possible.
Scalability adapts only to the "workload increase" by "provisioning" the resources in an "incremental" manner.	Elasticity adapts to both the "workload increase" as well as "workload decrease" by "provisioning and deprovisioning" resources in an "autonomic" manner.
Scalability enables a corporate to meet expected demands for services with "long-term, strategic needs".	Elasticity enables a corporate to meet unexpected changes in the demand for services with "short-term, tactical needs".

Deployment (serve in various locations)



Deployment (**serve in various locations)**

- Deployment Prototyping can help in making deployment architecture design choices by comparing performance of alternative deployment architectures.
- Life cycle of Cloud application deployment consists of **three phases :**

1. Deployment Design

- In this phase the application deployment is created with various **tiers** as specified in the deployment configuration.
- The components in this phase include the number of servers in each tier, computing, memory and storage capacities of servers, server interconnection, load balancing and replication stages.
- The process of resource provisioning and deployment creation is often automated and involves a number of steps such as launching of server instances, configuration of servers and deployment of various tiers of the application on the servers.

Deployment

2. Performance Evaluation

- Second Phase in the deployment life cycle is to verify whether the application meets the performance requirements with the deployment.
- In this phase involves **monitoring the workload** on the application and measuring **various workload parameters** like response time, throughput, utilization of servers in each tier.

3. Deployment Refinement

- After evaluating the performance of the application, deployments are refined so that the application can meet the performance requirements.
- Various alternatives can exists in this phase like vertical scaling(scaling up), Horizontal scaling(Scaling out),alternative server interconnections, alternative load balancing and replication strategies.

Examples for Popular Cloud Deployment Management Tools are RightScale, Scalr, Kaavo, Cloud Stack

Deployment

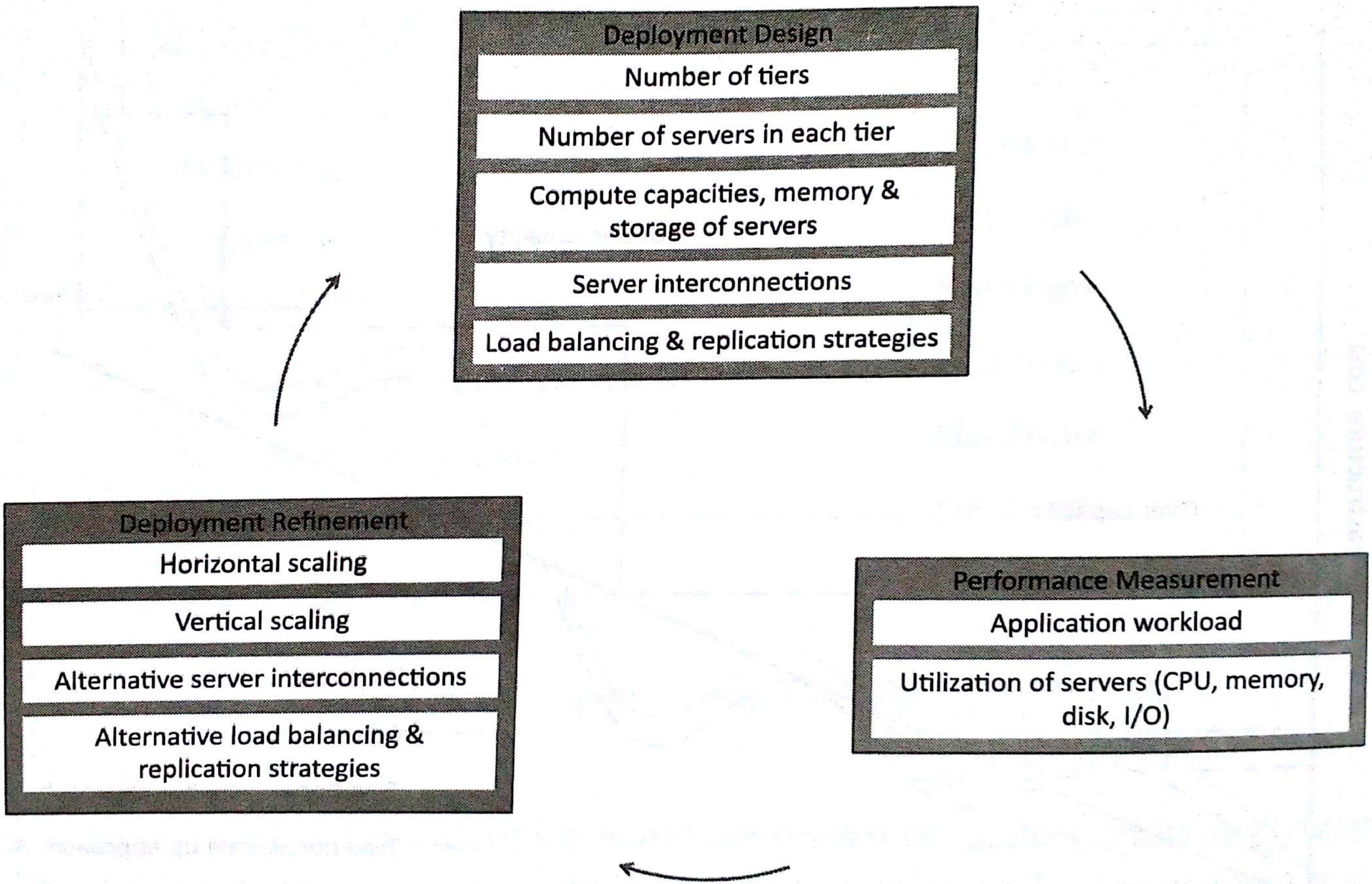


Fig : Cloud application Deployment lifecycle

Replication (reproducing something)

- It is a **method of creating and maintaining multiple copies of the data in the cloud.**
- It is important for the practical reasons like business continuity and disaster recovery.
- Cloud based data replication approaches provide replication of data in multiple locations, automated recovery, low **recovery point objective** (RPO) and **low recovery time objective**(RTO).
- RPO is the **maximum targeted period in which data might be lost from an IT service due to a major incident.**
- RTO is **defined as the loss of service time due to an incident.**
- There are three type of replication approaches :

1. **Array-based Replication**
2. **Network- based Replication**
3. **Host-based Replication**

Replication (reproducing something)

1 . Array-Based Replication

- Array-based replication uses compatible storage arrays to **automatically copy data from a local storage** array to a remote storage array.
- Arrays replicate data at sub-system level, therefore the type of hosts accessing the data and the type of data is not important.
- So that array-based replication can work in heterogeneous environments with different operating system.
- It uses Network Attached Storage (NAS) or Storage Area Networks(SAN) to replicate.
- Main drawback of this array-based replication is that it requires similar arrays at local and remote locations.

Replication (reproducing something)

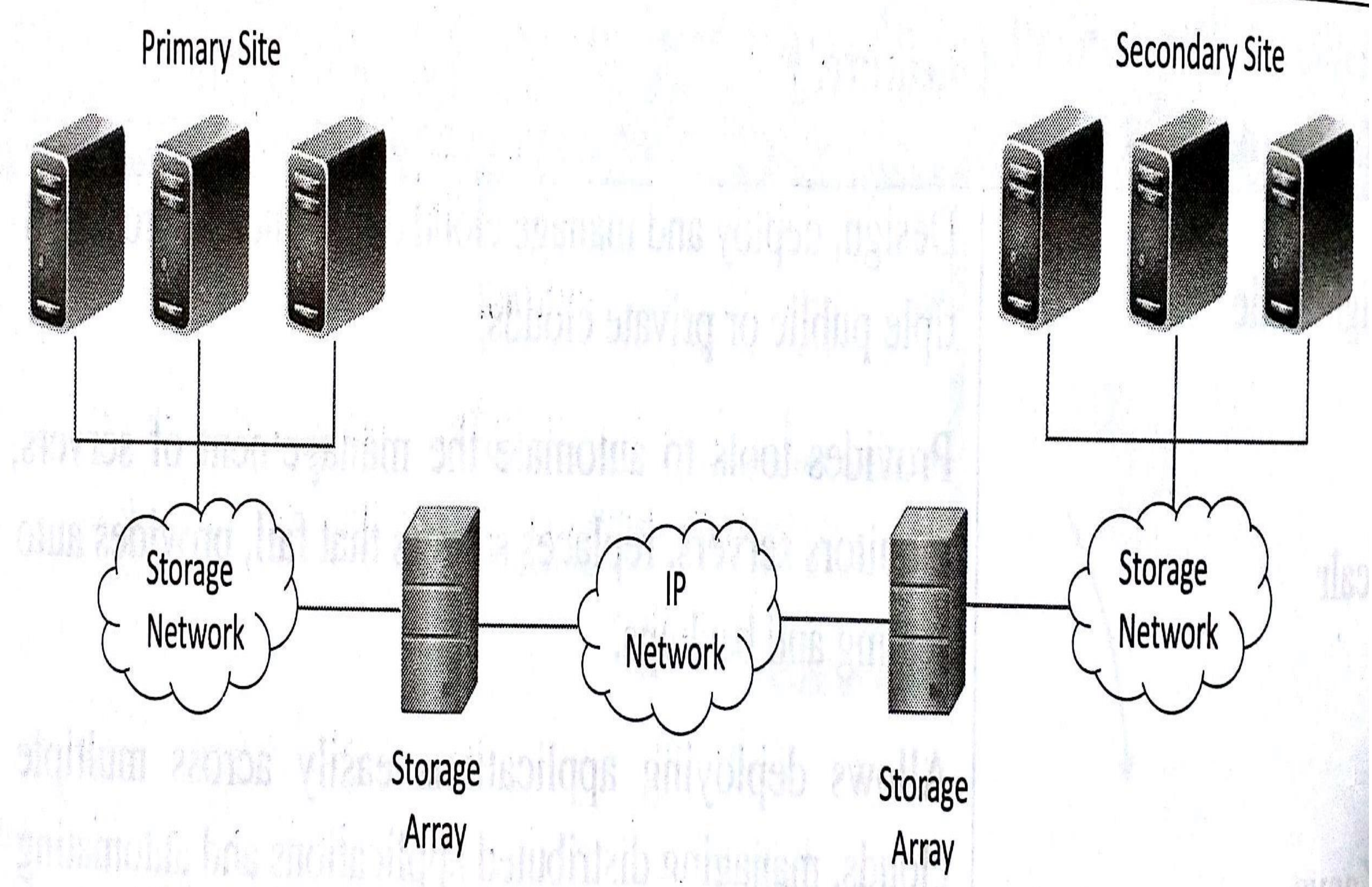
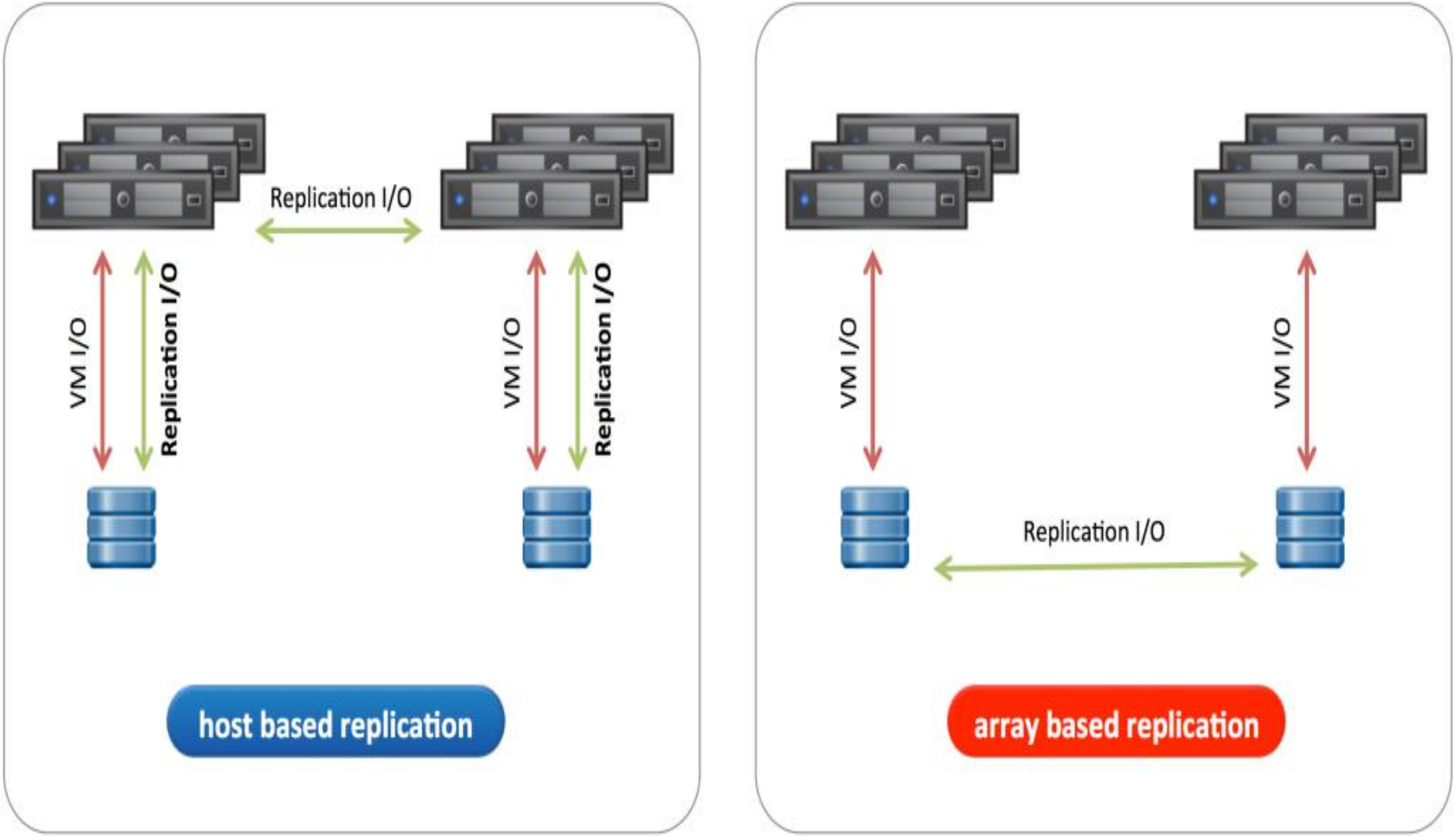


Fig : Array-based Replication



Replication I/O
VM I/O
Replication I/O
VM I/O
Replication I/O
host based replication
VM I/O
VM I/O
Replication I/O
array based replication

Replication (**reproducing something**)

2. Network-based Replication

- Network-based replication uses an **appliance** (**piece of equipment designed to perform a specific task.)** that sits on the network and intercepts packets that are sent from host and storage arrays.
- The intercepted packets are replicated to a secondary location.
- **Main advantage** of this approach is that it supports heterogeneous environments and requires a single point of management.
- **Drawback** of this approach involves higher initial costs due to replication hardware and software.

Replication

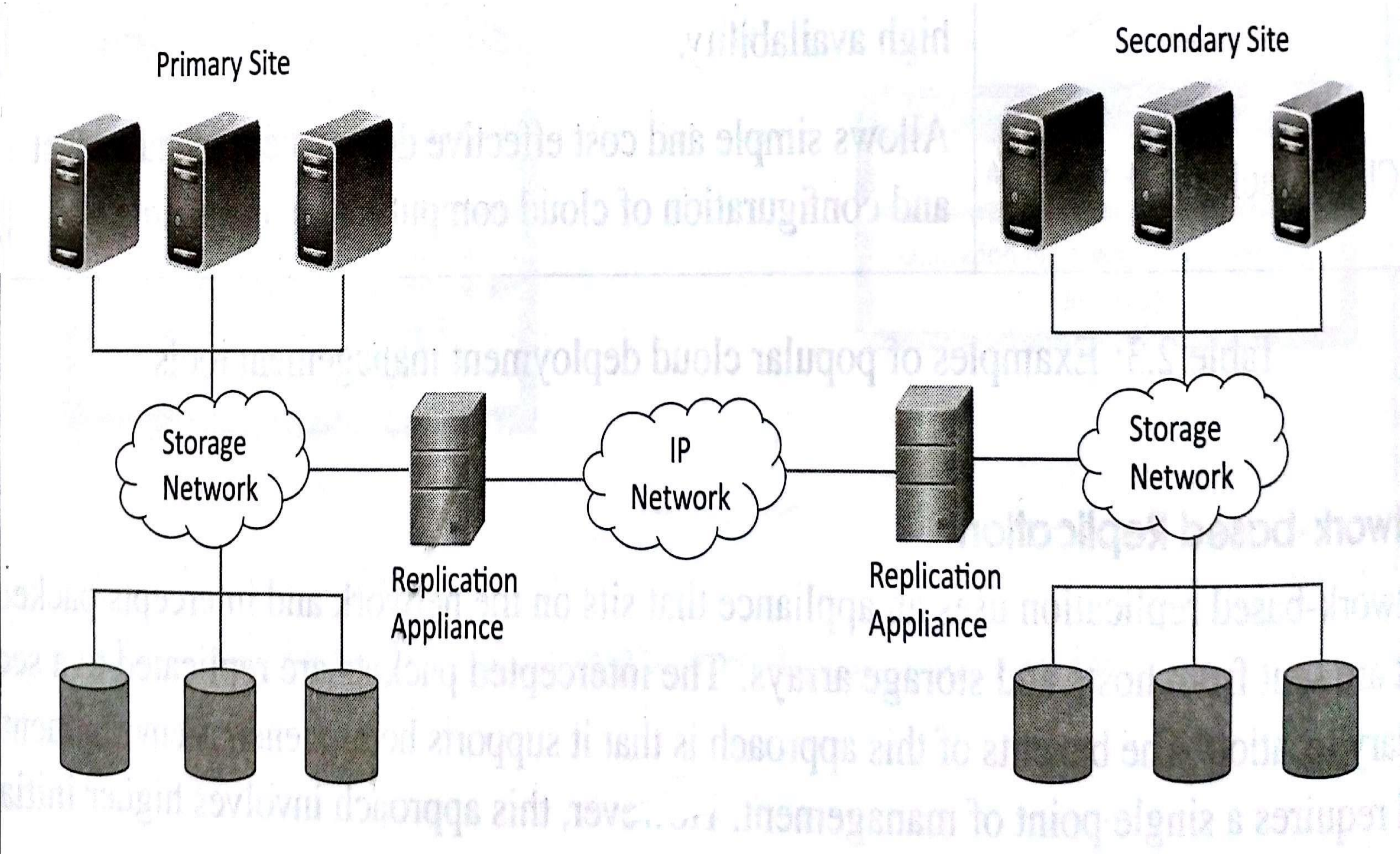


Fig : Network-based Replication

Replication (reproducing something)

3. Host-based Replication

- It Runs on standard servers and uses software to transfer data from local to remote location.
- The hosts acts the replication mechanism.
- An agent is installed on the hosts that communicate with the agents on the other hosts.

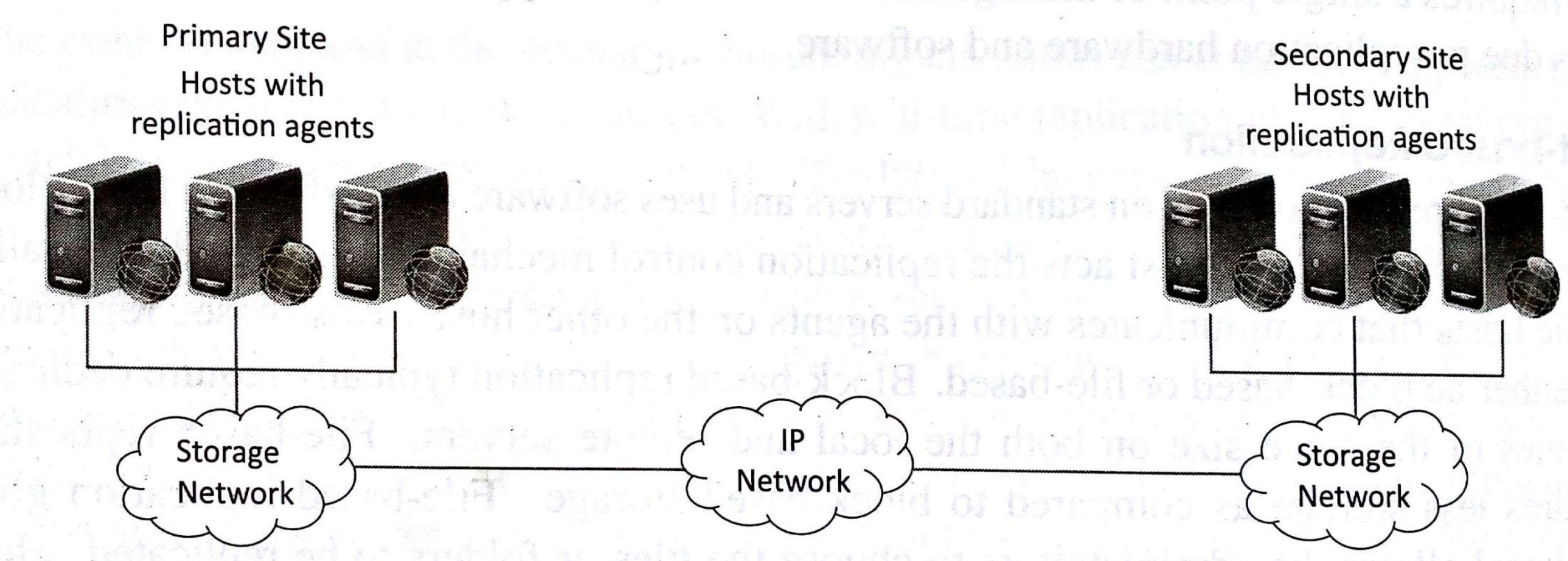


Fig : Host-based Replication

- When creating remote replicas for business continuity, the decision to deploy a host- or a storage-based solution depends heavily on the platform being replicated and the business requirements for the application(s) in the event of a disaster. If the business demands no impact to operation in the event of site disaster, host-based replication techniques provide the only feasible solution.

Monitoring

- Cloud Resources can be monitored by monitoring services provided by the **cloud service providers**.
- Monitoring services allow cloud users to collect and analyze the data on various monitoring **metrics** (provides knowledge about a **cloud)** **like CPU, Disk, Memory and Interface.**
- Monitoring service collect data on various system and application metrics from the cloud computing instances.
- Monitoring of cloud resources is important because it allows the user to **keep track of the health of applications and services deployed in the cloud.**

Typical monitoring metrics

Type	Metrics (provides knowledge about a cloud)
CPU	CPU-Usage,CPU-Idle
Disk	Disk-Usage,Bytes/sec(read/write),Operations/Sec
Memory	Memory-Used,Memory-Free,Page-Cache
Interface	Packets/sec,Octets/sec(incoming/outgoing)

Monitoring

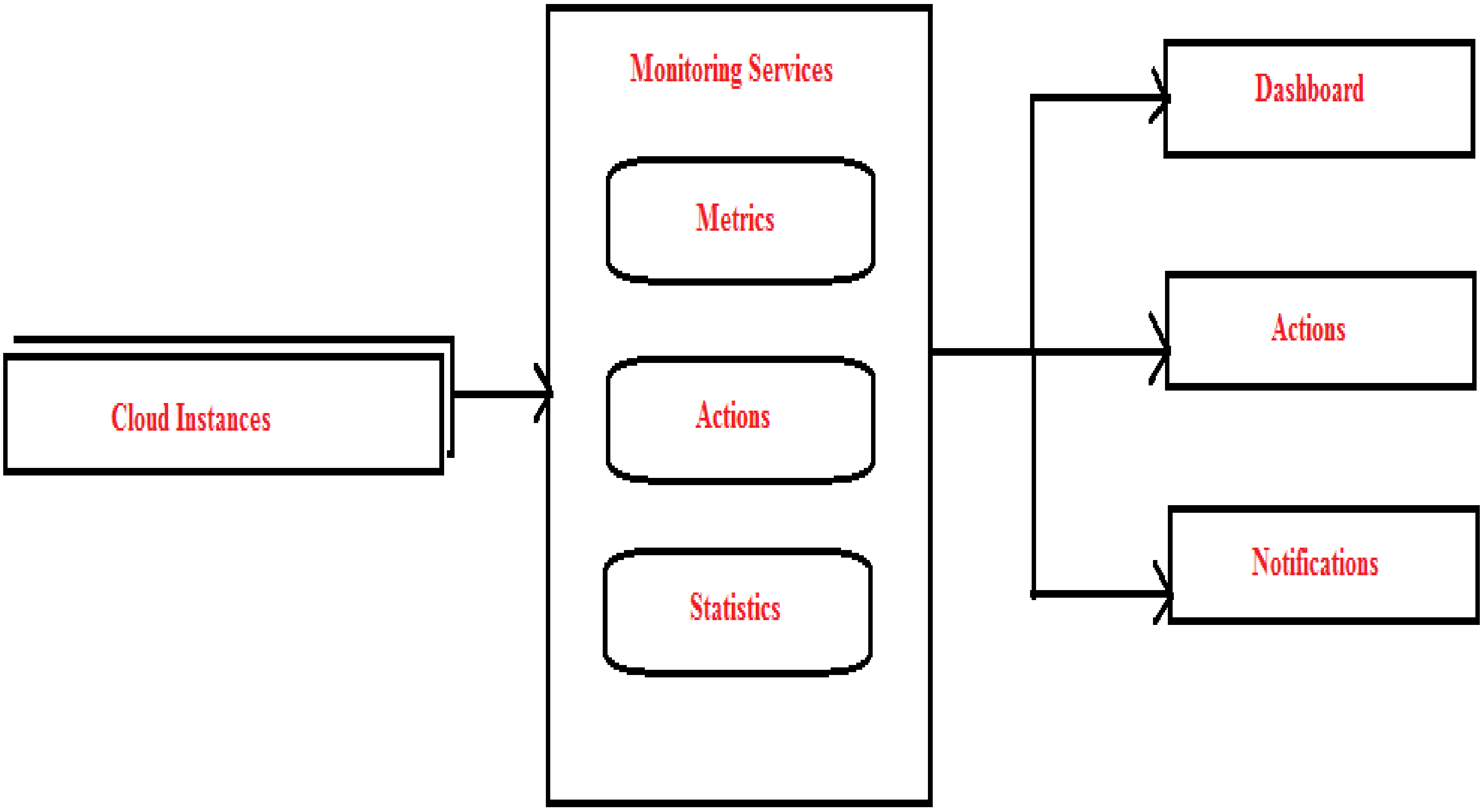


Fig : Monitoring Services Architecture

MapReduce

- Map Reduce is a **parallel data processing model for processing and analysis of massive (Large) scale data**.
- It has two phases : **1. Map 2. Reduce**
- **Input data to the map** and **reduce functions** in the form of key value pairs.
- **In the Map phase**, data is read from a distributed file system, partitioned among a set of computing nodes in the cluster and set to the nodes as a set of key-value pairs.
- The intermediate results are stored on the local disk of the node running the map task.
- **In Reduce phase**, the intermediate data with the same key is aggregated**(gather)** .
- MapReduce programs take the advantage of locality of data and the data processing takes place on the nodes where the data resides.

MapReduce

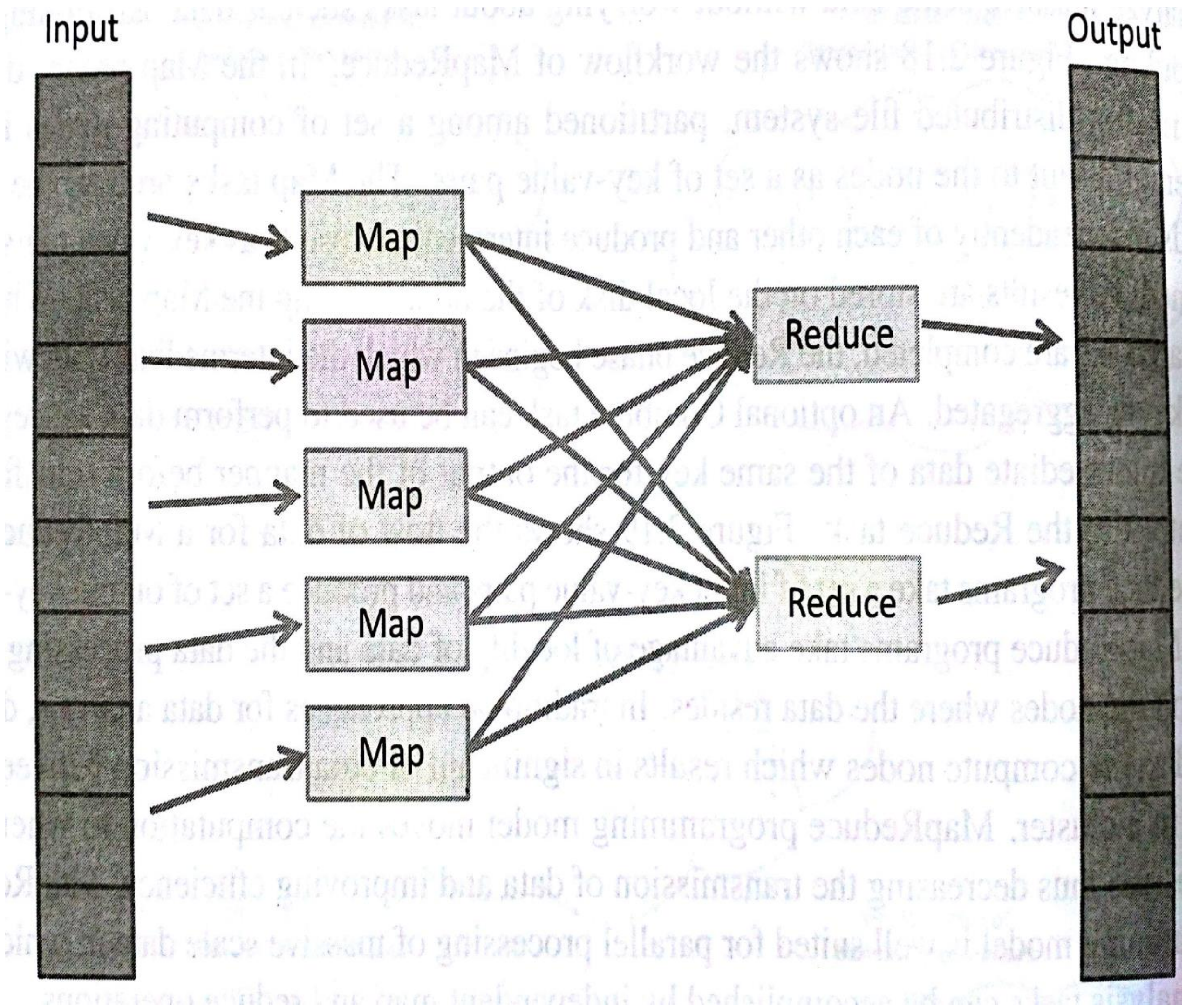


Fig : Dataflow in MapReduce

MapReduce

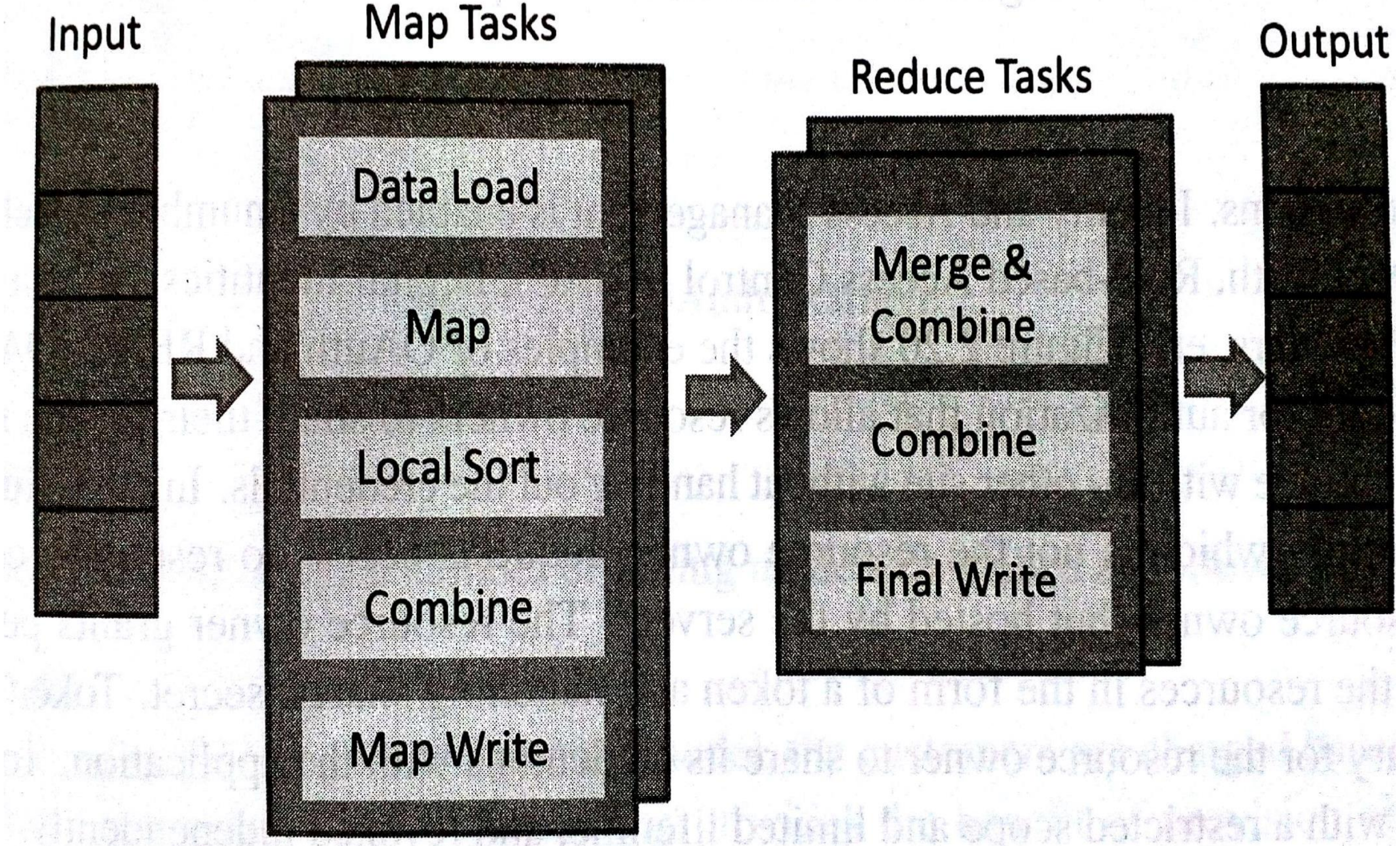
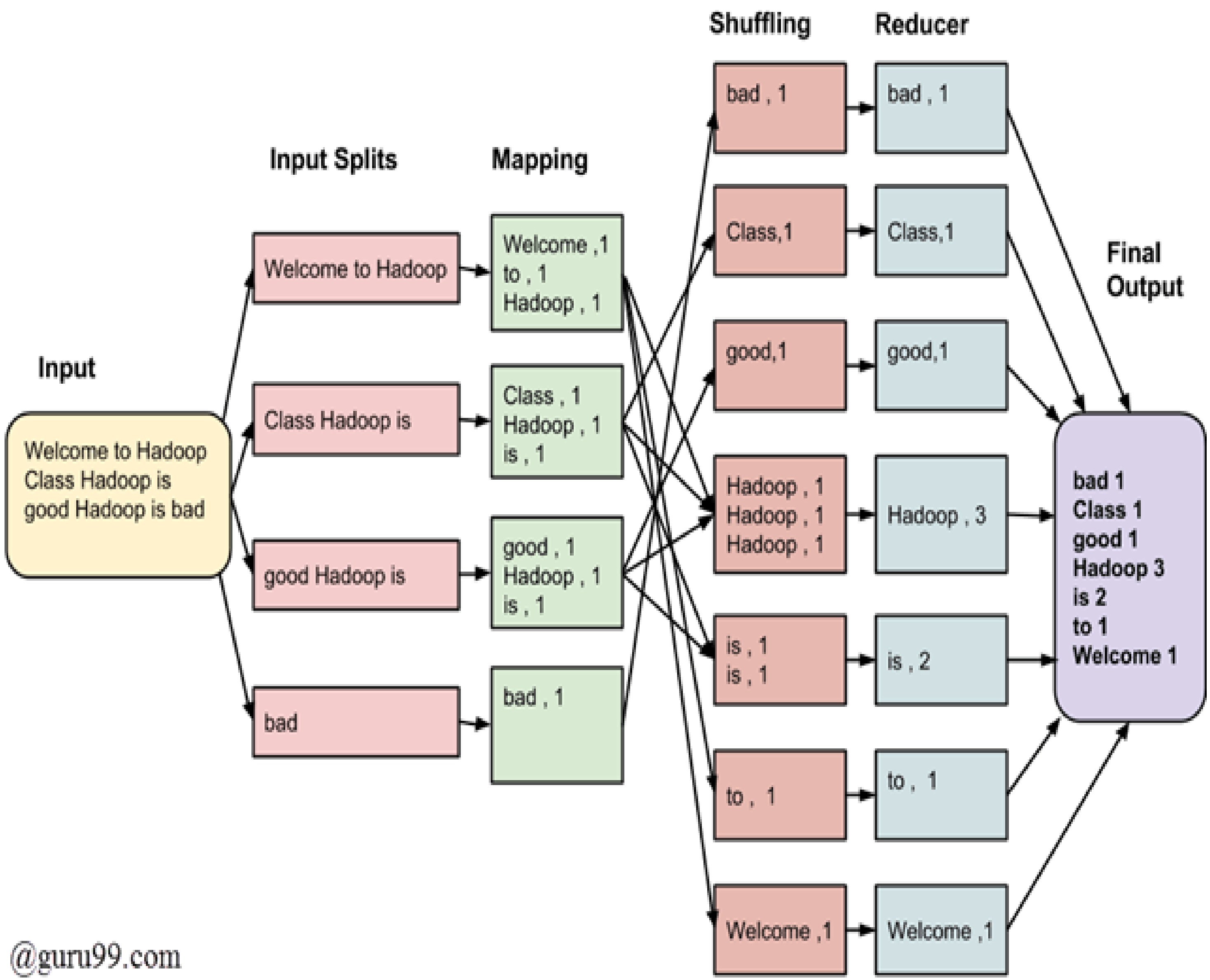
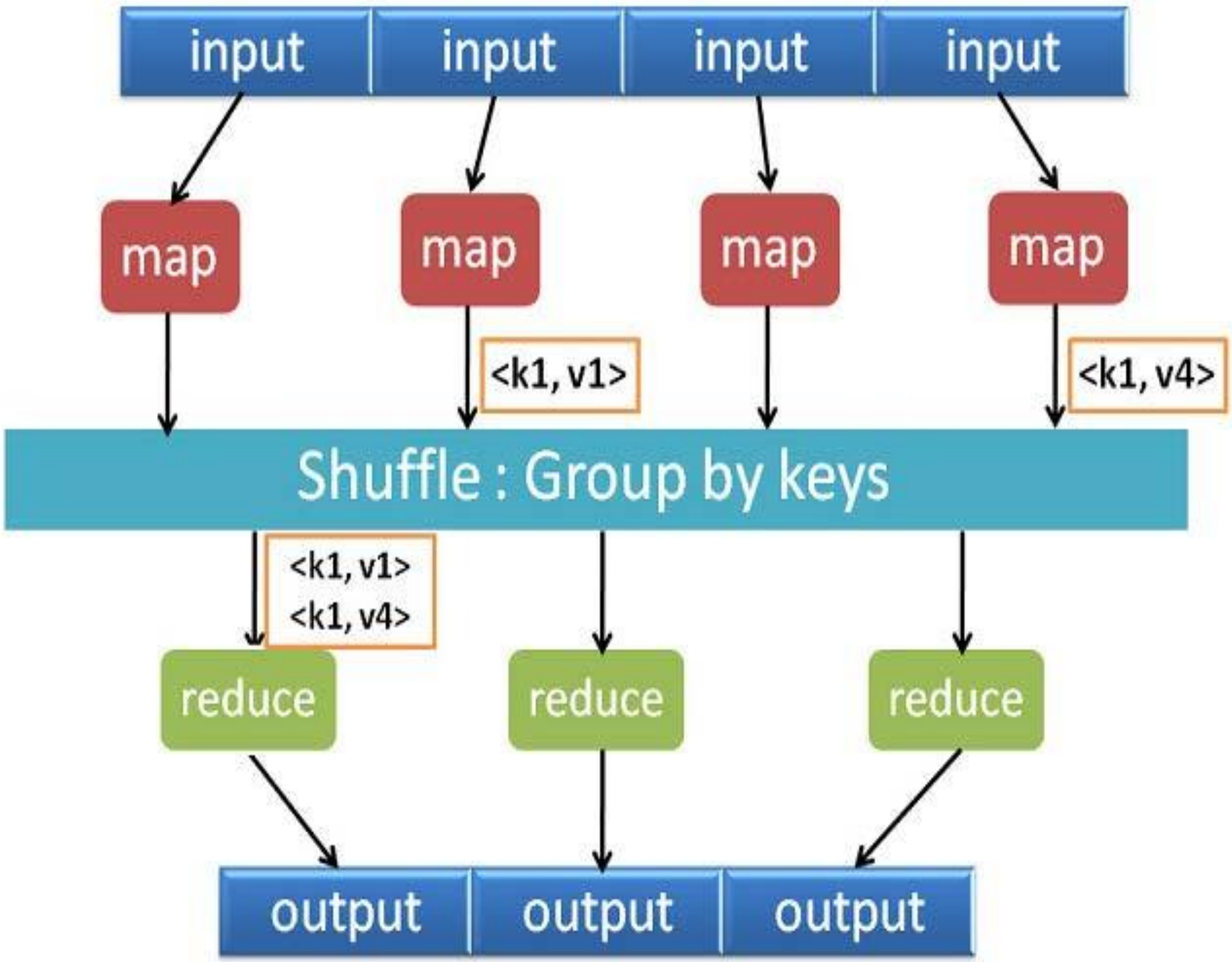


Fig : MapReduce Work flow

MapReduce

- The MapReduce runtime system take care of tasks such partitioning the data, scheduling of jobs, and communication between nodes in the cluster.
- This makes the programmers easy to analyze massive scale data without worrying about tasks such as data partitioning and scheduling.
- In traditional approaches of data analysis, data is moved to the compute nodes which results in significant of data transmission between the nodes in a cluster.
- In case of MapReduce model moves the computation to where the data resides so that it reduces the transmission of data and improves efficiency.


Input
Welcome to Hadoop Class Hadoop is good Hadoop is bad
Input Splits
Welcome to Hadoop
Class Hadoop is
good Hadoop is
bad
Mapping
Welcome ,1
to , 1
Hadoop , 1
Class , 1
Hadoop , 1
is , 1
good , 1
Hadoop , 1
is , 1
bad , 1
Shuffling
bad , 1
Class,1
good,1
Hadoop , 1
Hadoop , 1
Hadoop , 1
is , 1
is , 1
to , 1
Welcome ,1
Reducer
bad , 1
Class,1
good,1
Hadoop , 3
is , 2
to , 1
Welcome ,1
Final Output
bad 1
Class 1
good 1
Hadoop 3
is 2
to 1
Welcome 1
@guru99.com


input
input
input
input
map
map
map
map
<k1, v1>
<k1, v4>
Shuffle : Group by keys
<k1, v1>
<k1, v4>
reduce
reduce
reduce
output
output
output

- **Shuffle phase** in Hadoop transfers the map output from Mapper to a Reducer in MapReduce.
- **Sort phase** in MapReduce covers the merging and sorting of map outputs.
- Data from the mapper are grouped by the key, split among reducers and sorted by the key.
- Every reducer obtains all values associated with the same key.
- Shuffle and sort phase in Hadoop occur simultaneously and are done by the MapReduce framework.

Identity and Access Management

- **Identity and Access Management(IDAM)** for cloud describes the authentication and authorization of users to provide secure access to cloud resources.
- IDAM allows organizations to centrally manage users, access permissions, security credentials and access keys.
- IDAM is enabled by a number of technologies such as OpenAuth(Oauth), Rolebased Access Control(RBAC), Digital Identities, Security Tokens etc.
- Organizations can enable role-based access control to cloud resources and applications using the IDAM services.
- IDAM services allow creation of user groups where all the users in a group have the same access permissions.

Identity and Access Management

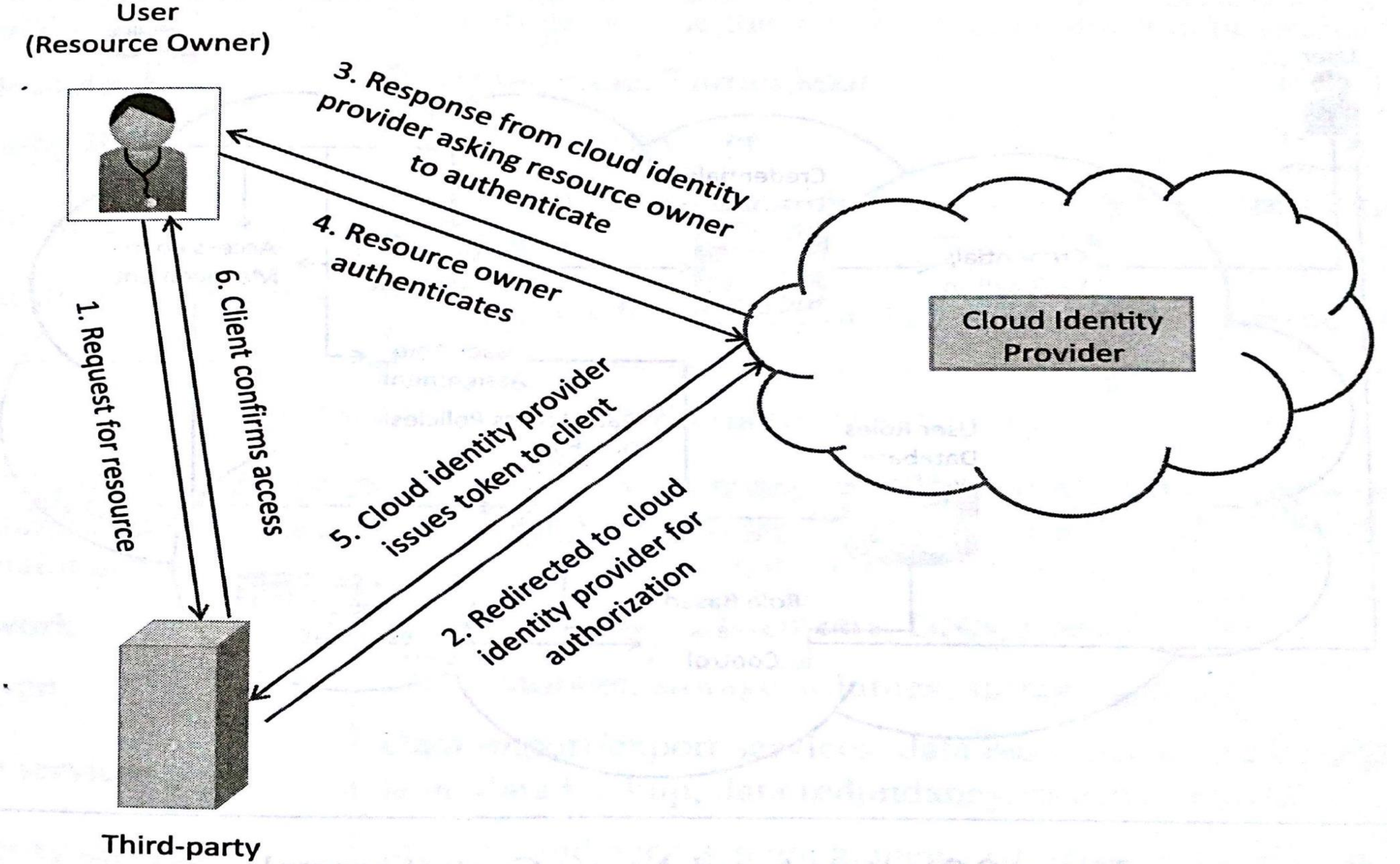


Fig : OAuth Example

Service Level Agreements

- A service level Agreement for cloud specifies the level of service that is formally defined as a part of the service contract with the service cloud provider.
- SLA provides a level of minimum level of service guaranteed and a target level.
- General cloud SLA criteria's are **Availability, Performance, Disaster Recovery, Problem Resolution, Security and Privacy of Data.**

Billing

- **Cloud security providers (CSP)** offer a different billing models like Elastic Pricing, Fixed Pricing, Spot Pricing.
- Main billable resources for cloud are like virtual machines, network, storage, data services, security services, support, application Services, Deployment and Management services.


SLA
Service-level agreement

List of Criteria for Cloud SLAs

Criteria	Details
Availability	Percentage of time the service is guaranteed to be available
Performance	Response time, throughput
Disaster Recovery	Mean time to recover
Problem Resolution	Process to identify problems, support options, resolution expectations
Security and privacy of data	Mechanisms for security of data in storage and transmission

List of billable resources for cloud

Resource	Details
Virtual machines	CPU,memory,storage,disk I/O,Network I/O
Network	Network I/O,load balancers,DNS,firewall,VPN
Storage	Cloud storage,storage volumes,storage gateway
Data Services	Data import/export services,data encryption,data compression,data backup,data redundancy ,content delivery
Security services	Identity and access management,isolation,compliance
Support	Level of support ,SLA,fault toleranc
Application services	Queuing service,notification service,workflow service,payment service
Deployment and management services	Monitoring service,deployment service

Thank You